

DAC on Padframe

In Lab 2, the purpose is to embed the 5-bit R-2R DAC that was built in Lab 1 in a padframe, which serves to provide connection to the outside world. The padframe is symmetrical in shape and is made using the least number of pad cells needed for the exported terminals of DAC. The DAC icon exposes six external signals that include b0, b1, b2, b3, b4 and vout. The ground is still an internal connection in the schematic of DAC and the final padframe has a total of six physical pads.

The design of pad cell was completed first that is used in a hierarchical fashion. Its layout provides the physical spot for bonding, as well as a Metal-3 terminal for the connection. Its schematic exports a single port called pin.

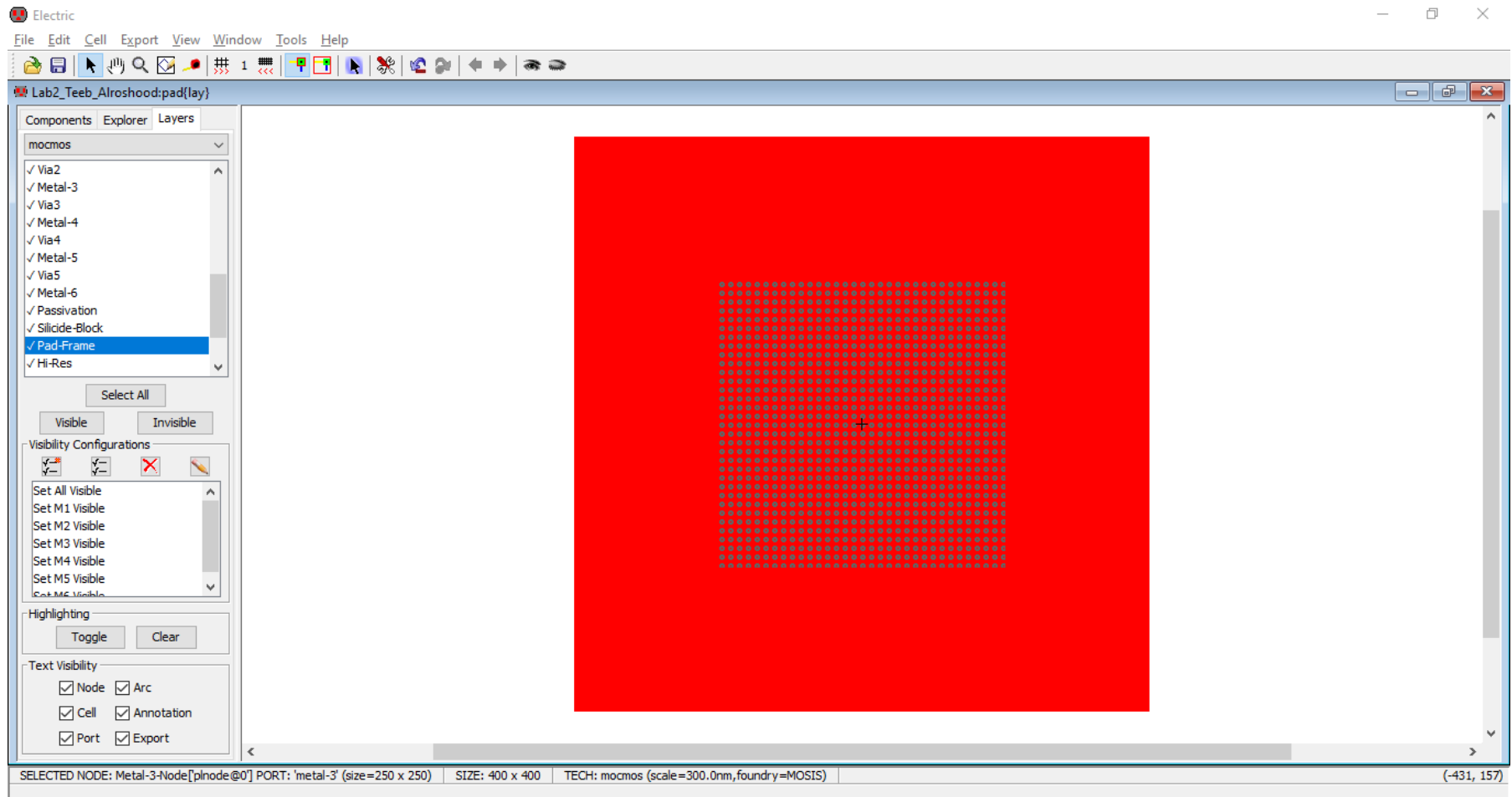


Figure 1. Electric layout of the reusable pad cell. The large pad-frame region surrounds the central Metal-3 connection area.

Pad cell schematic

Pin represents the exported terminal in the diagram of the pad cell which has its interface at a single port leading to ease of manufacturing the pad cell as an array and facilitating a uniform internal structure in all pads which leads to same pad cell in schematic and layout forms.

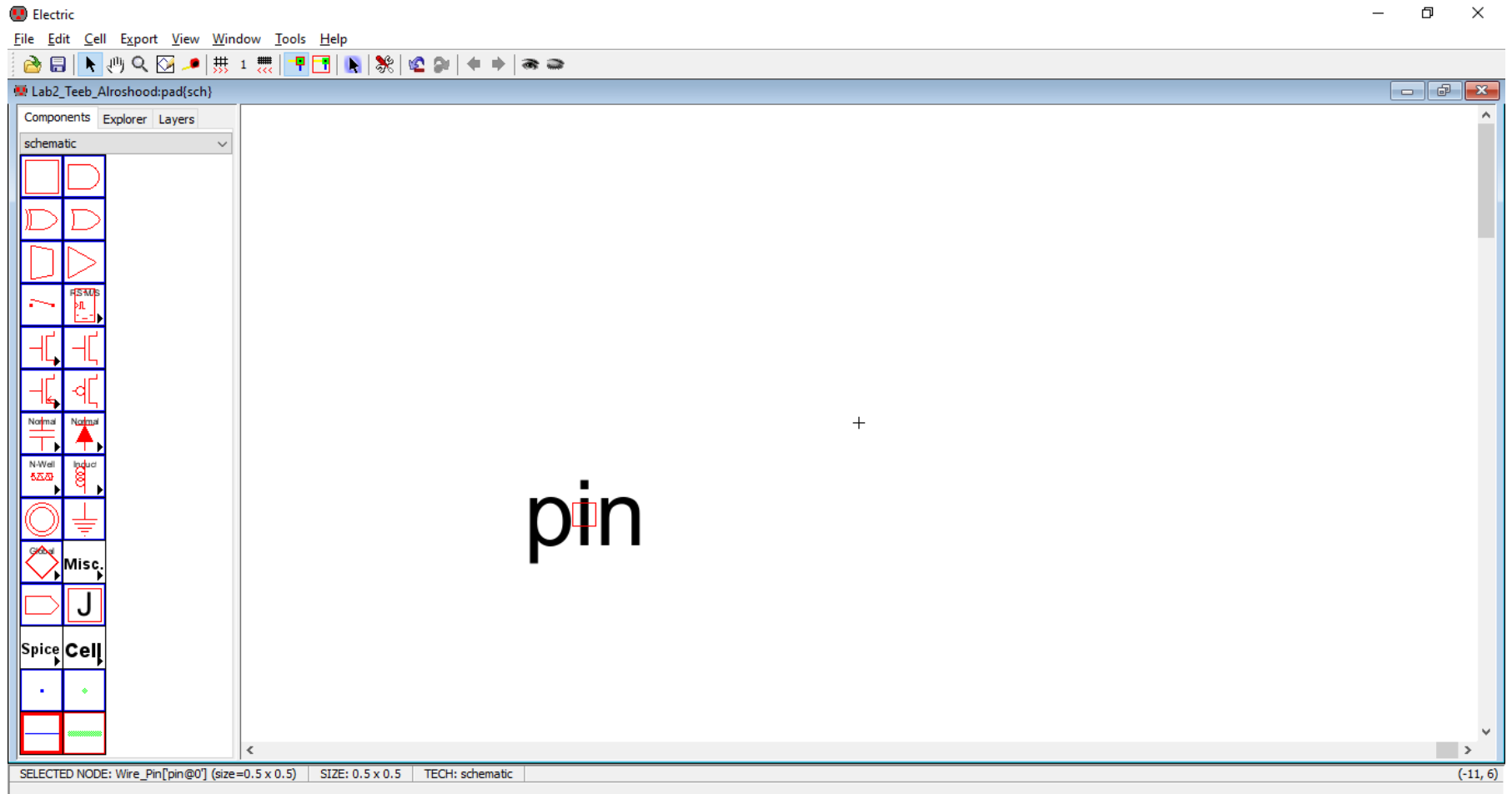


Figure 2. Pad-cell schematic showing the exported pin terminal.

DAC schematic reused from Lab 1

The Lab 1 digital-to-analog converter is a 5-bit R-2R ladder built using 16 resistors of 10 kOhm each. The five digital data inputs, labeled b4 to b0, control the ladder. For Lab 2, this cell is used hierarchically without having to redraw it. As a result, the only external inputs will be the five digital input bits and the analog output.

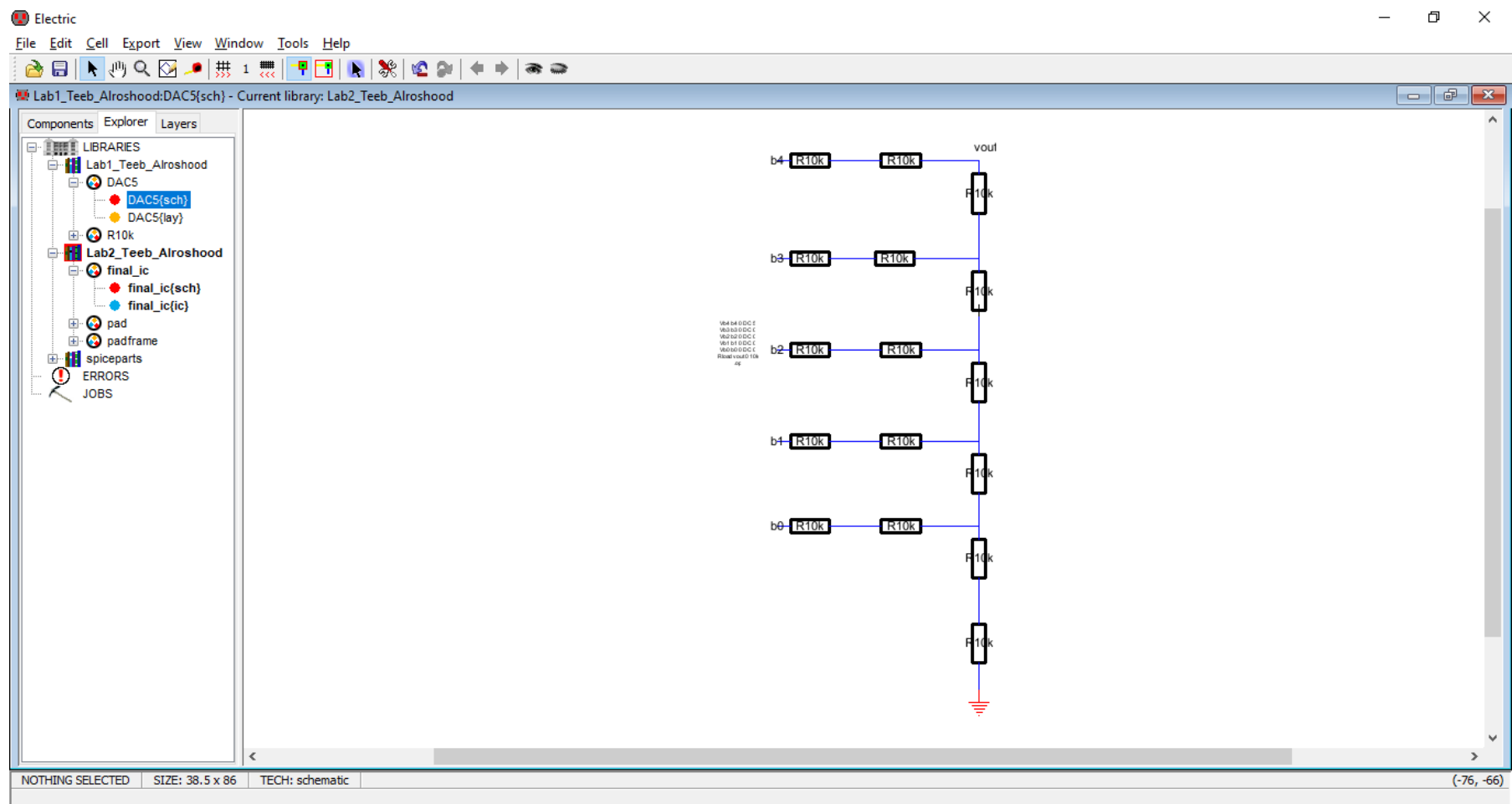


Figure 3. Schematic of the 5-bit R-2R DAC reused from Lab 1.

Padframe schematic construction

The tutorial shows how to create the padframe using repeated instances of the pad cell. The following screenshot shows the vector array construction stage during the padframe schematic compilation. The final completed design only needed six active pad nets because the DAC contained six external pins. Thus, the Lab 2 condition of employing the least number of pads to connect all DAC outputs was met.

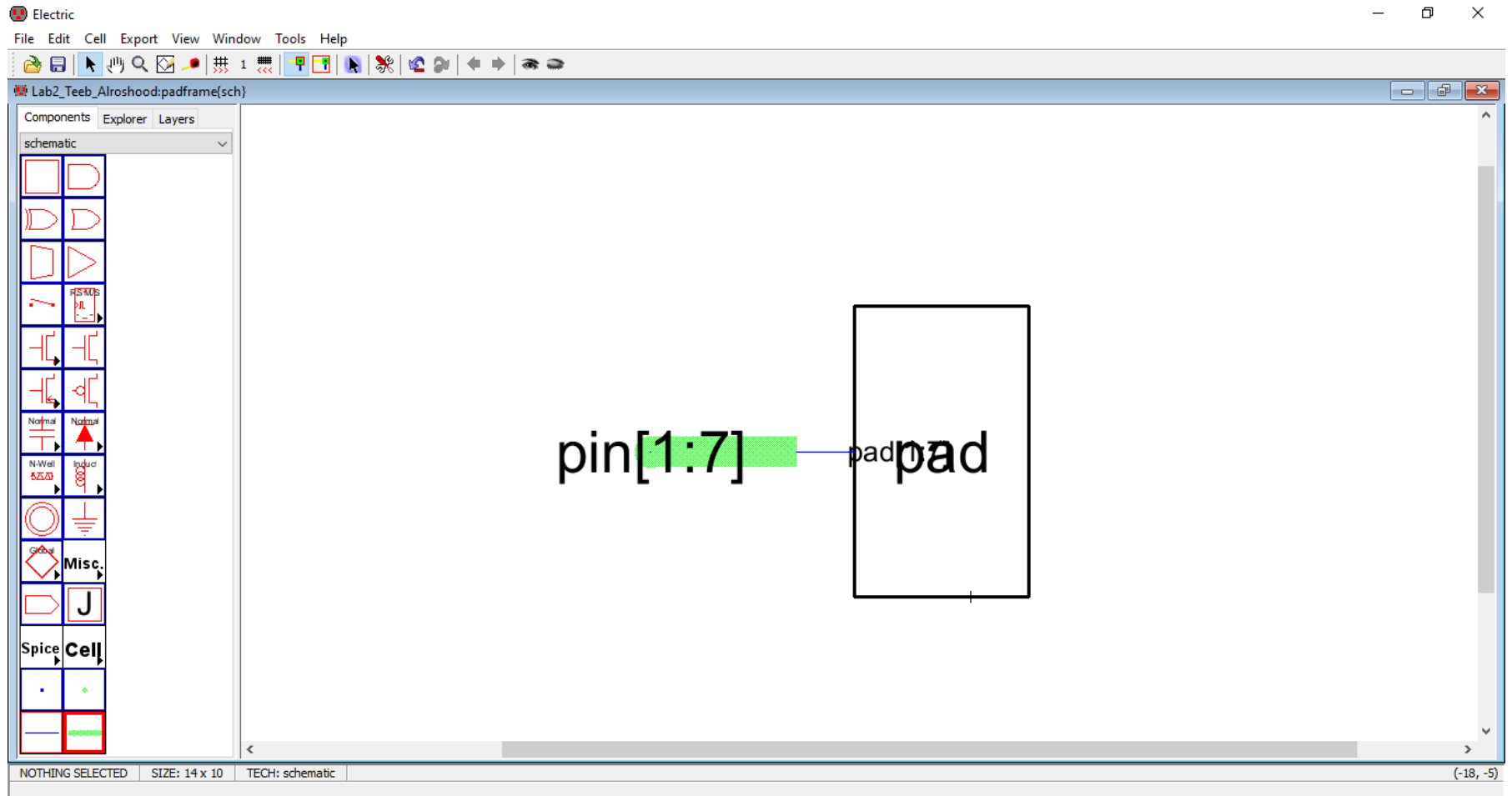


Figure 4. Padframe schematic during vectorized pad-cell construction. The final integrated interface uses six active pad connections.

Final IC schematic

In the final_ic schematic diagram, all of the six DAC pins are connected to each of the six padframe nets. The assignments of pins are as follows: pin[1] – b0, pin[2] – b1, pin[3] – b2, pin[4] – b3, pin[5] – b4, pin[6] – vout. Using the pins with identical numbers in the schematic and layout ensures the clear hierarchy and makes each physical connection easily understandable.

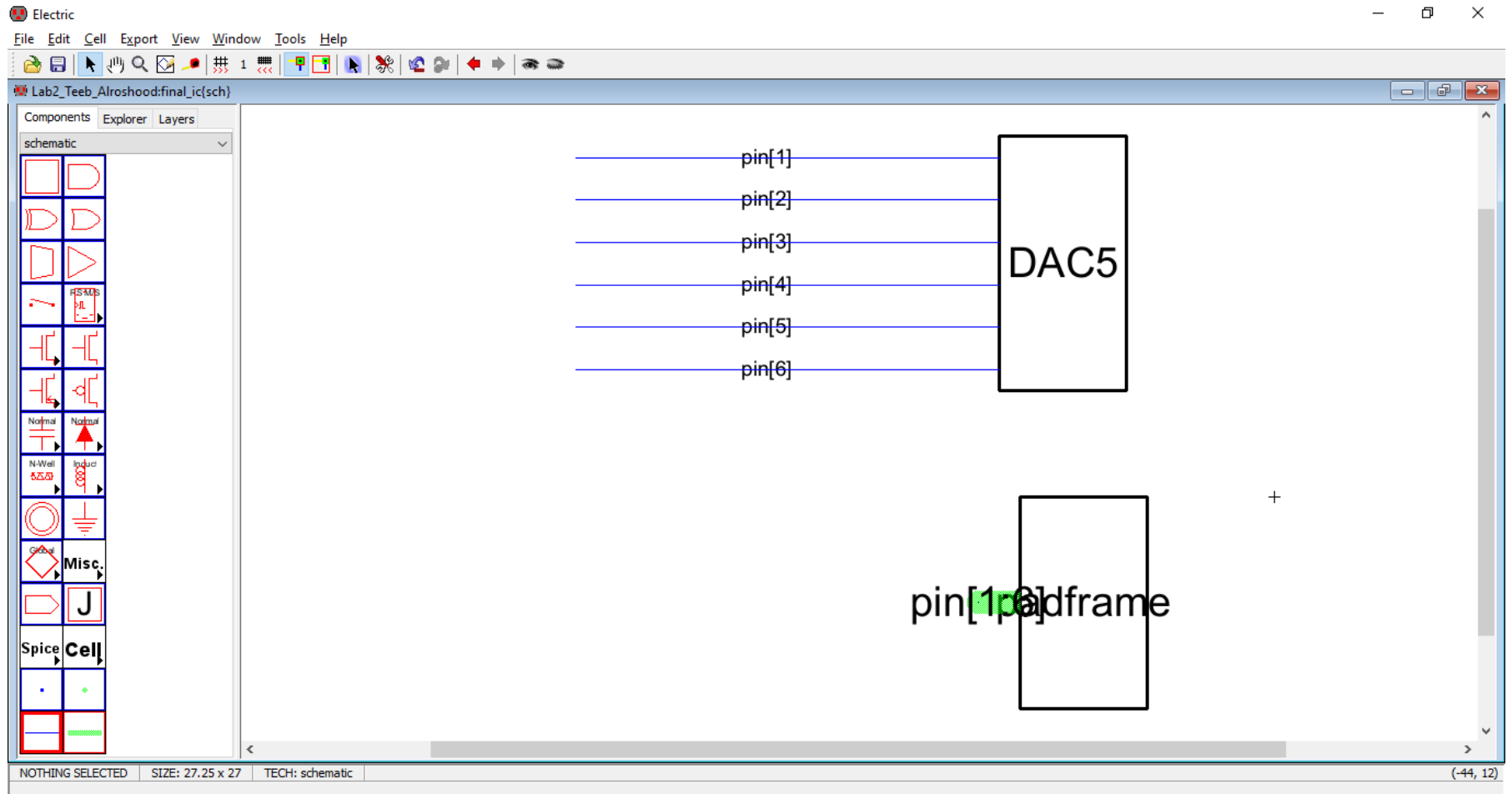
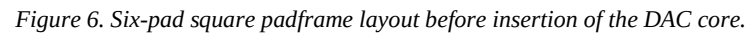


Figure 5. Final final_ic schematic with the six DAC connections and the padframe interface.

Six pad-cell layout instances have been arranged around the vacant central area in such a way that the total outline remains around a square while keeping the area for DAC intact. Two pads are put at the top, two at the bottom, and one each on the sides. This method utilizes the least amount of pads while positioning the external connections on the border of the die.



Final IC layout and routing

The DAC layout was found at the center of the padframe, and the six DAC terminals were laid out connecting to the respective pads. The DAC output is available at Metal-1 whereas the pad can be contacted at Metal-3. Hence, the Metal-1-Metal-2 and Metal-2-Metal-3 contacts were applied where a layer transition occurred. The routing did not interfere with any DAC Metal-1 features and the Metal-3 paths were ended up at expressing pads.

Routing performed kept the schematic connection layout pin[1] to pin[6] intact and contained a clear center where the pads were located.

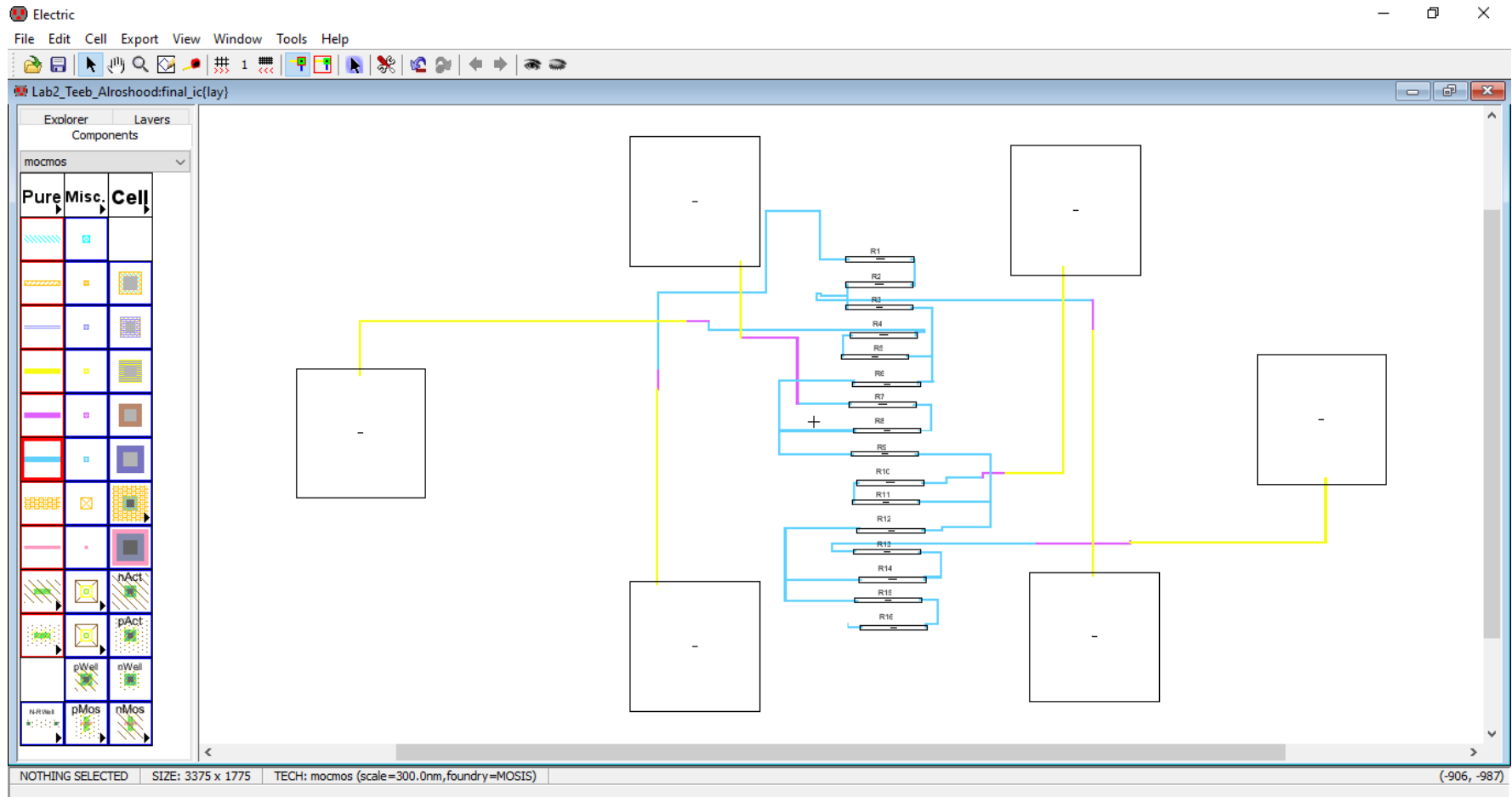


Figure 7. Integrated final_ic layout with the DAC centered inside the six-pad frame and all six external routes completed.

Verification

The design checked was the hierarchical layout using the DRC flow of Electric Design Automation Tools. Metal-1 notches/spacings and Metal-3 pad overlaps were corrected while routing the layout by relocating layer transition points, using non-metal-1 shapes, and making sure that Metal-3 arcs end on the output pad ports and not just overlapping the pad geometry. A check was run after the routing corrections by going for Tools > DRC > Check Hierarchically.

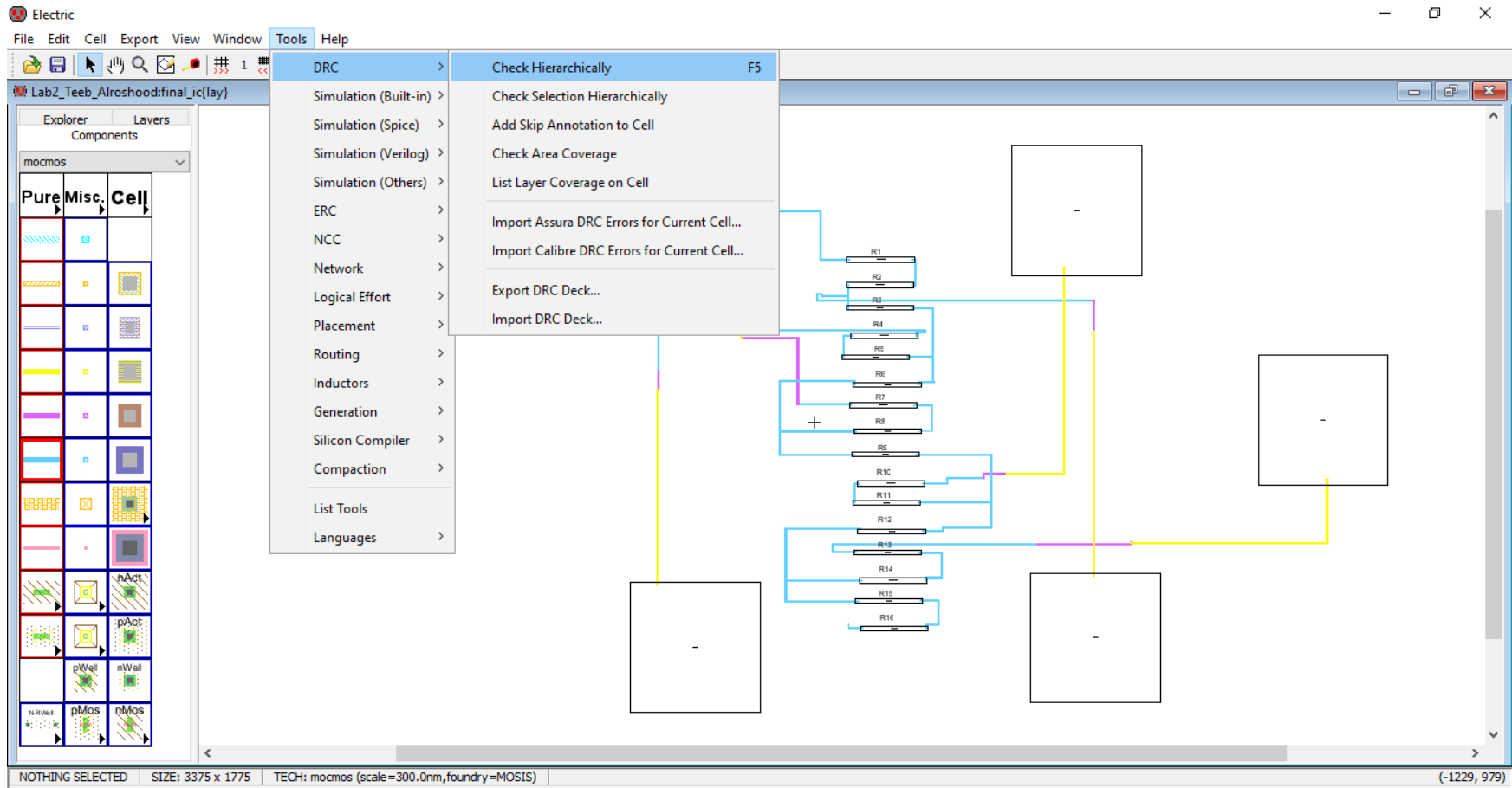


Figure 8. Electric hierarchical DRC command used for final layout verification.

Conclusion

The 5-bit Digital to Analog Converter of Lab 1 was assembled into a square case having six-contact pads using a total of five digital inputs and one analog output only. The required hierarchy includes the reusable pad cell, the pad frame, the final_ic schematic, and the routed final_ic layout, when the Metal-Co terminals of the digital-analog converter are connected through the necessary layers.